

NCA Assistant

User Manual

Version 1.6

Non-Compartmental Pharmacokinetic Analysis

Bioequivalence • Drug Interaction Studies • Power & Sample Size

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Document Control

Version	Date	Author	Changes
1.0	March 2026		Initial release
1.1	April 2026		Added theoretical background (Part V), expanded statistical methods (Part VI), DDI study analysis, decimal separator guidance
1.2	April 2026		Quick Start data preparation callout, grid plot fix documentation, single-subject Analysis Record, DDI interpretation thresholds, regulatory DDI guideline links, troubleshooting updates
1.3	April 2026		New chapter 3 (Visualize Data); chapters 3–5 renumbered to 4–6; hub layout updated to 3+3 grid; attribution updated
1.4	April 2026		Factual corrections (Rule 6 label, AUC extrapolation flag); new feature documentation (steady-state in BE, balance check, design summary, Apply LLOQ button, excluded profiles note); beginner clarity additions (LLOQ/BLQ definitions, extravascular explained, R^2 failure consequences, SHA-256 explained, half-life review how-to, AUC % Extrapolated meaning)
1.5	September 2026		Updated for NCA Assistant v1.4.0: CDISC ADNCA upload and parameter codes, safety checks on upload, decimal comma and semicolon files, doses per period, replicate designs and the within-subject variability table, choosing the Reference treatment, the minimum R^2 rule, planning with Test and Reference CV, the reproduction check in the Analysis Record. Exercise 1 answers corrected; guideline references updated.
1.6	September 2026		Updated for NCA Assistant v1.5.0: partial AUCs in Paths 4, 5 and 6, with intervals ending at a time or at the last measurable concentration, Cmax and Tmax within an

			interval, pivotal and supportive metrics in bioequivalence, and interval shading in Visualize Data. New Exercise 4, new sections in Parts V and VI, and new glossary, troubleshooting and reference entries.
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Update: right-click → Update Field → Update Entire Table.

Quick Start Guide

Before you begin: check that your data file has the right format. If this is your first time, read Part II (Preparing Your Data) first; most upload errors come from formatting.

Step 1. Open the app at `robterheine.shinyapps.io/NCAassistant` or run `source("install_and_run.R")` in RStudio.

Step 2. Click Upload & Check Data. Choose what kind of file you have (a simple table, or a CDISC ADNCA dataset), upload the CSV or Excel file and check the column mapping the app suggests.

Step 3. If the data contain BLQ values, set the LLOQ and choose a BLQ rule. When the app finds entries such as "<0.5", it offers a blue "Apply LLOQ = X and process" button. Otherwise enter the LLOQ yourself and click Process Data. Leave the LLOQ at 0 if there are no BLQ values.

Step 4. Optionally, open Visualize Data to look at the individual profiles and the geometric mean curve first. Then go to All Subjects (Batch), set dose, units and route, and click Run Analysis on All Subjects.

Step 5. Check the results. Download them (Excel or CSV) or generate a Complete Analysis Record.

For bioequivalence or drug interaction studies, use Bioequivalence (Path 6). Plan a Study needs no data. The Data Preparation Guide, with example datasets, is on the home screen and in the top navigation bar.

Part I: Introduction to Non-Compartmental Analysis

What Is Non-Compartmental Analysis?

Non-compartmental analysis (NCA) is a method for describing the pharmacokinetics of a drug without assuming a specific structural model for how the drug distributes in the body. Instead of fitting a mathematical model, NCA calculates pharmacokinetic parameters directly from the observed concentration–time data using numerical integration and regression.

NCA is the standard approach for pharmacokinetic data in drug development. EMA, FDA and PMDA use it for bioequivalence studies, and it is routine in Phase I trials, dose-finding studies and therapeutic drug monitoring research.

The Concentration–Time Curve

After a drug is administered, blood samples are collected at specified times and the drug concentration is measured. When plotted against time, the curve shows absorption (concentrations rising), distribution (transition), and elimination (concentrations declining). NCA extracts quantitative parameters from this curve without assuming its shape.

Key Pharmacokinetic Parameters

Parameter	Symbol	What it tells you
Peak concentration	C _{max}	The highest concentration observed. Relevant for efficacy and safety.
Time of peak	T _{max}	When the peak occurs. Reflects the rate of absorption.
Area under the curve	AUC	Total drug exposure over time. The most important measure of overall exposure.
Terminal half-life	t _{1/2}	Time for the concentration to halve during the terminal phase.
Apparent clearance	CL/F	Volume of plasma cleared of drug per unit time, divided by the bioavailability F.
Volume of distribution	V _z /F	Apparent volume the drug distributes into during the terminal phase.
Mean residence time	MRT	Average time a drug molecule stays in the body.
Partial AUC	AUC t ₁ –t ₂	Exposure over part of the profile, for example the first 30 minutes. It catches a difference in the shape of the curve that C _{max} and total AUC can miss.

Why NCA Is Used So Widely

NCA does not assume a compartmental structure, so the same calculations apply whether a drug follows one-compartment kinetics or multi-phasic elimination. Results depend on the observed data and the numerical method, not on model selection, which makes them reproducible across software and analysts.

Regulatory agencies accept NCA because the methods are well-defined and transparent. NCA is the required approach for bioequivalence studies, where C_{max} (rate) and AUC (extent) are the primary endpoints.

When to Use NCA vs. Compartmental Modelling

Use NCA when you need standard PK parameters from observed data: first-in-human studies, bioequivalence, drug interaction studies, dose proportionality, food effect studies.

Use compartmental modelling when you need to simulate unobserved scenarios, when you have sparse sampling, or when you need to understand mechanisms. The two approaches are complementary.

Part II: Preparing Your Data

General Format Rules

The app expects data in **long format**: one row per sample, with a subject identifier, the sampling time and the measured concentration. Accepted files are CSV (.csv), text (.txt, .tsv) and Excel (.xlsx, .xls). The app suggests a column mapping from the column names (Appendix B); check it before you process the data. CDISC ADNCA datasets are uploaded differently, see Scenario 9.

Decimal Separators and Regional Formats

Choose the delimiter (comma, semicolon or tab) and the decimal point (period or comma) in the upload settings. With a decimal comma, values such as 12,5 are read correctly, also in a concentration column that contains BLQ text such as “<0,5”. Excel files need no settings, because Excel stores numbers as numbers.

Use one concentration unit, one time unit and one dose unit for the whole file, and select them in the app. They set the conversion factor for clearance and volume, so a wrong unit gives wrong CL/F and Vz/F while Cmax and AUC look normal. A unit column with more than one unit (even ng/mL and µg/L) is refused; convert the values first. Molar concentrations (for example nmol/L) with a mass dose (mg) need the molecular weight, which you enter in the analysis settings.

Scenario 1: Single Subject

To upload a file, include a Subject column; it may hold the same value on every row. Without it the upload cannot tell which rows belong together, and the app picks another column as Subject. Manual entry in One Subject at a Time needs only Time and Concentration.

Subject	Time	Concentration
P1	0	0
P1	0.5	4.2
P1	1	18.5
P1	2	12.1
P1	4	6.3
P1	8	2.8
P1	24	0.4

Scenario 2: Multiple Subjects, Same Dose

Subject	Time	Concentration
S001	0	0
S001	1	22.5
S001	24	0.4
S002	0	0
S002	1	28.1
S002	24	0.6

Scenario 3: Multiple Subjects, Different Doses

Add a Dose column if subjects received different doses (dose escalation, weight-based dosing), or if a subject received different doses in different periods. The app takes the dose of each profile (subject, treatment and period) from this column, so every period is analysed with its own dose.

Scenario 4: Standard 2-Period Crossover (BE)

Six columns: Subject, Treatment, Period, Sequence, Time, Concentration. Time restarts at 0 in each period: it is the time since the dose of that period, not since the first dose of the study (the app refuses profiles that do not start near time zero). The ratio is Test ÷ Reference. In Bioequivalence you choose which treatment is the Reference; the app pre-selects it only when the name is unambiguous (R, Ref, Reference, Comparator, Innovator, Originator or RLD).

If the protocol asks for a partial AUC, take a sample at every cutoff time. A cutoff between two samples is interpolated, so its value then depends on the trapezoidal method. The interval itself comes from the protocol or the product-specific guidance, never from the data.

Subject	Treatment	Period	Seq	Time	Conc
S01	Test	1	TR	0	0
S01	Test	1	TR	1	24.5
S01	Reference	2	TR	0	0
S01	Reference	2	TR	1	22.8

Scenario 5: Replicate Designs and Paired Comparison

Same six-column format. Replicate designs: 2×2×3 full replicate (sequences TRT and RTR), 2×3×3 partial replicate (TRR, RTR, RRT) and 2×2×4 full replicate (TRTR, RTRT). Period goes to 3 or 4 and a treatment appears more than once per subject; each administration is analysed as its own profile. Paired comparison: all subjects received the treatments in the same order, so Sequence has one value (or can be left out).

Scenario 6: Parallel-Group BE

Four columns: Subject, Treatment, Time, Concentration. No Period or Sequence needed.

Scenario 7: Steady-State PK

Same layout as Scenario 2, with one dosing interval per subject. Time 0 is just before the dose of that interval, and the pre-dose concentration is above zero. Include the pre-dose sample and sample up to the end of the dosing interval (τ). Tick “Steady-state (drug given repeatedly)” and enter τ in the app.

A partial AUC at steady state has to lie within 0 and τ . AUC_{τ} is completed to τ with λ_z when the last sample comes earlier; a partial AUC you enter yourself is not, so it stays empty for such a profile.

Scenario 8: Drug–Drug Interaction Study

A drug interaction (DDI) study measures the effect of a co-administered drug (the perpetrator) on the pharmacokinetics of the victim drug. Most are crossovers in which each subject receives the victim drug alone and with the perpetrator, or fixed-order studies in which the victim is given alone first and with the perpetrator after a washout.

Use the same six columns as a crossover BE study (Scenario 4). Label the victim drug alone as “Reference” and the victim drug with the perpetrator as “Test”, or choose the Reference treatment in the app. The ratio is then (with perpetrator) ÷ (alone), the fold-change in exposure. Part III explains how to interpret the result.

Scenario 9: CDISC ADNCA Dataset

If your data come as a CDISC ADNCA dataset (columns such as USUBJID, AVAL, PARAMCD, NRRLT, ARRLT and ANL01FL), do not convert them by hand. On the Upload page, set “What kind of file?” to “CDISC ADNCA dataset” and upload the .csv or .xlsx file. The app shows a summary and asks which time to use (NRRLT nominal, ARRLT or MRRLT actual; AFRLT, time since the first dose, is refused) and, if there are several, which analyte. It applies ANL01FL, drops records with PCSTAT “NOT DONE”, and refuses datasets it cannot convert safely, such as derived records (DTYPE), several units, or AVAL missing without a BLQ result. The choices are stored in the Analysis Record. SAS transport (.xpt) files are not read: converters/ADNCA_TO_FLAT.md in the repository gives a two-line R recipe to convert them to CSV. The same conversion is available outside the app as converters/adnca_to_flat.R.

BLQ Handling

Rule	Description
Rule 1	Pre-first-quantifiable = 0, post-last-quantifiable = missing, between = 0 (default)
Rule 2	All BLQ = 0
Rule 3	All BLQ = excluded
Rule 4	All BLQ = LLOQ/2
Rule 5	Before Cmax = 0, after Cmax = missing
Rule 6	Before first quantifiable = LLOQ/2, rest = 0 (for drugs with absorption lag)

Every assay has a lower limit of quantification (LLOQ): the lowest concentration it can measure reliably. Samples below it are reported as BLQ (below the limit of quantification). The app recognises “<0.5” (also “<0,5”), BLQ, BQL, BLOQ, ND (not detected) and NQ (not quantifiable), in any capitals. Set the LLOQ above 0 so the chosen rule is applied to these entries; each rule works per profile, in time order. NS (no sample), N/A and “missing” are treated as missing values, not as BLQ. When the app finds “<X” entries, it offers an “Apply LLOQ = X and process” button.

In the processed data the app adds a BLQ_flag column marking every value the rule set. It appears in the data preview, and the partial AUC notes and the bioequivalence table use it to report how much of a result rests on such values. A column of that name in your file is replaced.

The rule matters most for partial AUCs. An early window often holds BLQ samples, which Rule 1 sets to 0 and Rule 6 to LLOQ/2, and a late window over a plateau near the LLOQ can rest mainly on values the rule produced. The app says so when more than half of the

samples in a window come from the BLQ rule, or when fewer than three measurable concentrations are left in it.

Common Mistakes

Wide format: subjects as columns instead of rows. **Time not reset:** in a crossover, time restarts at 0 in every period. **0 for a missing sample:** leave the cell empty, or write NS. **Duplicate time points:** usually several analytes, matrices or periods in one column; filter the file, or map the Treatment and Period columns. **LLOQ left at 0:** processing is blocked when BLQ text is present. **Dates or clock times as time:** the Time column must be elapsed time since the dose. **Mixed or wrong units:** select the units that match the file; they drive the clearance and volume conversion.

A cutoff with no sample near it: the protocol asks for AUC 0–30 min, but the samples are at 15 and 60 min. The app interpolates the concentration at 30 min, so the value depends on the trapezoidal method.

Part III: Using the App

The Hub Screen

Six cards for the six workflow paths, in two rows of three: Plan a Study, Upload & Check Data and Visualize Data on top; One Subject at a Time, All Subjects (Batch) and Bioequivalence below. The top navigation bar links to Statistical Methods, the Data Guide, About (with the version history) and this User Manual.

Path 1: Plan a Study

Calculates the sample size, or the power for a given number of subjects, with the PowerTOST package. Choose the method: standard average bioequivalence (ABE), highly variable drugs (EMA ABEL or FDA RSABE), or narrow therapeutic index drugs (FDA NTID). The designs are those of Bioequivalence except the paired comparison: 2×2 , $2 \times 2 \times 3$, $2 \times 3 \times 3$, $2 \times 2 \times 4$ and parallel groups for ABE; the replicate designs for ABEL and RSABE; only $2 \times 2 \times 4$ for NTID.

Enter the expected Test/Reference ratio (the default follows the method, as in PowerTOST: 95% for standard bioequivalence, 90% for highly variable drugs, 97.5% for narrow therapeutic index drugs), the CV and, for sample size, the target power. For a parallel design the CV is the total CV (between plus within subjects); for crossover designs it is the within-subject CV. For the scaled methods enter two CVs, of the Test and of the Reference product; both change the sample size. The significance level (0.05) and, for ABE, the acceptance limits can be changed under the advanced settings. After a log-transformed bioequivalence analysis of your own data, a button offers its within-subject CV for C_{max}. The spread of C_{max} between subjects in batch results is not a within-subject CV, so the app does not offer it. The results show a power curve and a plot of sample size against CV.

For a pivotal partial AUC, plan with the within-subject CV of that partial AUC. An early window is often more variable than total AUC, so a CV taken from AUC gives too few subjects.

Path 2: Upload & Check Data

Choose what kind of file you have, upload it, check the column mapping, set the LLOQ and BLQ rule, and click Process Data. The app runs more than 20 quality checks: errors block processing, warnings do not. Some checks stop data the app cannot analyse safely: a CDISC-style file uploaded as a simple table, mixed units, dates or clock times in the Time column, profiles that do not start near time zero (time since the first dose), and several profiles stacked in one column (duplicate times). Processing a new file clears all previous results. Afterwards a panel links to the next steps: Visualize Data, Single Subject, Batch Analysis and Bioequivalence.

Path 3: Visualize Data

Creates publication-ready concentration–time plots from uploaded data. Accessible directly from the hub or via the “Visualize Data” button on the Upload page. Requires data to be uploaded and processed first; a prompt is shown otherwise.

Layout: settings sidebar on the left, stacked plot area on the right. There are no tabs: the Plot Type control decides which plots appear.

Plot Types

The Plot Type radio button has three options. “Individual Profiles (spaghetti plot)” draws one concentration–time line per subject, coloured by a grouping of your choice; use it to look at individual variability, outliers and carry-over. “Summary Plot (geometric mean $\times/\div$ geometric SD)” shows the geometric mean at each time point, with error bars from the geometric mean divided and multiplied by the geometric SD. Treatment groups are overlaid automatically. “Both plots” shows the spaghetti plot above the summary plot.

Concentrations at a time point are usually skewed to the right, so the geometric mean describes the typical profile better than the arithmetic mean. Arithmetic mean $\pm$ SD is available, with a warning.

Sidebar Controls

Y-axis scale (Linear or Log₁₀): the semi-log view makes the terminal elimination phase linear, which is useful for identifying biphasic profiles or half-life estimation issues. When log scale is active and zero concentrations are present they are omitted from the plot and a note is shown.

Colour lines by (spaghetti plot, dropdown): the options come from the columns mapped during upload: Subject ID (always available), Treatment, Period, Sequence (each available only when that column was mapped), and None. In a crossover, colour by Period to look for period effects, or by Treatment to compare formulations.

Normalize to dose (show C/Dose), a checkbox that appears only when a Dose column was mapped: concentrations are divided by the dose, so the Y axis shows C/Dose (for example ng/mL per mg). Useful for dose-escalation studies. A missing or zero dose gives a missing value. Change the Y-axis label to match, for example “C/Dose (ng·mL⁻¹·mg⁻¹)”.

Colour palette: “Default (app palette)”, “Colorblind-safe (Okabe-Ito)”, or “Grayscale”. For manuscripts destined for print or journals that do not support colour, select Grayscale and export at 600 DPI.

Theme: “Clean (white background)” (recommended for reports), “Minimal”, or “Classic”. Axis labels, plot title, and base font size (8–18 pt) are editable text fields. The default Y-axis label is “Concentration”; edit it to add units, e.g. “Cyclosporin (ng/mL)”. The default X-axis label is “Time (h)”.

Suggested Figure Legend

Above the plot area, a card titled “Suggested figure legend” holds a ready-to-copy description of the current figure, generated from your data and settings: number of subjects and observations, plot type, summary statistic, error bars, treatment names, the number of concentrations ≤ 0 left out of the geometric mean, Y-axis scale and colour grouping. It updates when you change settings. Paste it into your report and add the drug name and units.

Exporting a Figure

Export controls are at the bottom of the settings sidebar under the heading “Export Figure”, always visible. A small “Will export” label above the Download button shows exactly which

plot will be downloaded. When “Both plots” is selected, a radio button appears asking which plot to export; only one file is produced per download.

Format: PNG (good for Word documents, presentations, and emails), PDF (vector; text and lines remain sharp at any zoom level; recommended for journal submission when the journal accepts PDF figures), or SVG (scalable vector graphics; open and edit in Inkscape, Adobe Illustrator, or similar). Width and height are set in inches (2–20 in each); typical journal requirements are 3.0–3.5 in for a single column and 6.5–7.0 in for double column.

Resolution: 150 DPI (screen/drafts), 300 DPI (print, default, minimum for most journals), or 600 DPI (high-resolution journal submission). The Download button is grey until the first plot has rendered in the main panel; it turns green once a plot is visible. The exported filename includes the plot type (spaghetti_plot.png or summary_plot.png) so you can download both without overwriting either.

Interactive Preview vs. Static Export

The plots shown in the main panel are interactive (built with plotly): hover over any point or line to see subject ID, time, and concentration; zoom by clicking and dragging; pan with the pan icon; reset with the home icon. These interactive features are not available in the exported static file. The static file (PNG/PDF/SVG) is built from a separate ggplot2 render using your sidebar settings (font size, theme and colour palette); use it for reports and manuscripts.

Worked Examples

Example 1: Quick data check after upload. Upload example_theoph.csv from the Data Preparation Guide. Navigate to “3. Visualize Data”. The individual profiles plot is shown by default: 12 coloured lines, one per subject, on a linear Y scale. Switch the Y axis to Log_{10} ; a straight tail points to mono-exponential elimination. This is a visual check only; no analysis settings are required.

Example 2: Summary plot for a crossover BE study. Upload a crossover dataset with a Treatment column mapped, open Visualize Data and select “Summary Plot”. The plot overlays one curve per treatment. This is a visual check, not a test: only the confidence interval in Bioequivalence tells whether the formulations are bioequivalent. For the report, go to Export Figure in the sidebar, select PDF, 7 × 5 in, 300 DPI, and click Download.

Example 3: Dose-escalation spaghetti plot. Upload data with a Dose column and different doses per subject. Tick “Normalize to dose (show C/Dose)”, colour the lines by Subject, and change the Y-axis label to “C/Dose ($\text{ng}\cdot\text{mL}^{-1}\cdot\text{mg}^{-1}$)”. With dose-proportional pharmacokinetics the normalised profiles overlap; profiles that diverge suggest non-linear pharmacokinetics.

Common Questions

The Download button is grey. The button is in the Export Figure section of the sidebar and activates once the first plot has rendered in the main panel. There is no separate tab to visit first; wait for the plot to appear.

Some time points are missing from my summary plot. The summary plot uses the nominal time values present in your data. If subjects have observations at slightly different actual times (e.g., 0.95 h vs 1.05 h for a nominal 1 h sample), round to nominal times before uploading. The app does not perform nominal-time imputation.

I see “N observations with concentration ≤ 0 excluded.” Zeros, for example BLQ values set to 0 by rules 1, 2 or 5, cannot be log-transformed, so they are left out of the geometric mean at that time point. They stay in your data.

My spaghetti plot has too many colours to distinguish. Try the Okabe-Ito (“Colorblind-safe”) palette. With more than eight groups, consider switching to Grayscale and annotating the exported figure in a graphics application.

Path 4: One Subject at a Time

Step through the profiles with Previous/Next, or enter data by hand. Set dose (filled in from the Dose column when there is one), units, route and minimum R^2 , and click Run PK Analysis. The half-life inspector lets you choose the terminal-phase points yourself. The result belongs to the profile it was computed for: selecting another profile clears it. Downloads: CSV and a Complete Analysis Record.

Partial AUCs (optional): set how many intervals you need and enter the start and end of each, in the time unit of your data. Tick “End at the last measurable concentration (t)” for an interval that runs to the end of the profile, and “Also report C_{max} and T_{max} in this interval” when the protocol asks for the peak inside a window. An interval that reaches past the last measurable concentration stays empty, with a note saying so: partial AUCs are never extrapolated.

Path 5: All Subjects (Batch)

Settings: route, dose (one dose for all, or from the Dose column per subject and period), units, steady state (with the dosing interval τ), trapezoidal method and minimum R^2 . Result tabs: concentration–time profiles, parameter table (37 columns when expanded; AUC % Extrapolated above 20% is flagged in amber), summary statistics per treatment, an individual grid, and Half-Life Review. When the automatic terminal fit of a profile has an adjusted R^2 below the minimum (default 0.70), its half-life, AUC $_{\infty}$, CL/F, Vz/F and MRT are left empty and a note names the profile; select the points in Half-Life Review if you want a value. Profiles with fewer than 2 positive concentrations are left out, with an amber note listing them. A CDISC parameter codes panel lists the official code of each parameter. Downloads: CSV, Excel (with summary statistics per treatment and a sheet of CDISC codes) and a Complete Analysis Record.

Partial AUCs (optional): set how many intervals you need and enter the start and end of each, in the time unit of your data. Tick “End at the last measurable concentration (t)” for an interval that runs to the end of each profile, and “Also report C_{max} and T_{max} in this interval” when the protocol asks for the peak inside a window. The columns join the results table, the summary statistics and the downloads. An interval that reaches past the last measurable concentration of a profile stays empty for that profile, and a note names the profile and its last measurable time.

With dose normalisation switched on, the partial AUCs and the C_{max} within an interval are divided by the dose as well; the time of that maximum is not.

Path 6: Bioequivalence

Runs the NCA, then the ANOVA, then the confidence interval (90% by default) and a forest plot. Choose the design: 2×2 standard crossover, 2×2×3 or 2×2×4 full replicate, 2×3×3 partial replicate, parallel groups, or paired comparison. Choose the Reference treatment (the ratio is Test ÷ Reference); the run is blocked until one is chosen, and the results table shows the comparison, for example “New / Old”. The model has Sequence, Subject, Period and Treatment as factors. With fixed effects this is EMA Method A; the mixed model (random subject intercept) is EMA Method B, not the FDA model for replicate designs. The Model column says which was fitted. The verdict uses confidence limits rounded to two decimals. With limits wider than 80–125% the point estimate must also lie within 80.00–125.00%, unless you untick that (only for comparisons without such a rule, such as drug-interaction no-effect boundaries). Half-life gets a ratio with its confidence interval but no verdict, because it is not a bioequivalence endpoint; Tmax, untransformed parameters and paired comparisons also get no verdict. If all subjects received the treatments in the same order, the data are analysed as a paired comparison. For replicate designs a table shows the within-subject variability of the Reference (and of the Test when replicated) and the EMA widened limits it would imply, for information only: the app does not issue a scaled (ABEL or RSABE) verdict. A design summary and an amber warning for subjects missing a treatment appear after the NCA. For steady-state data tick “Steady-state (drug given repeatedly)” and enter τ ; AUC τ is then compared instead of AUC to last point and AUC $^\infty$.

Settings: route (oral/IM/SC or IV bolus; IV infusion is not supported here), dose, units, trapezoidal method, minimum R², the parameters to compare (Cmax, AUC to last point and AUC to infinity by default; Tmax and half-life can be added), log transformation, confidence level (80 to 99%) and acceptance limits. Tabs: the comparison with forest plot, the PK parameter table, the ANOVA tables, concentration-time profiles, individual profiles and Half-Life Review. Downloads: Excel, CSV and a Complete Analysis Record.

Partial AUC intervals here also carry a role. A pivotal interval receives a bioequivalence verdict, a supportive interval only its ratio and confidence interval. The metrics appear under “Parameters to compare” as soon as the intervals are valid, and Cmax within an interval can be compared as well. A metric whose value is zero in any profile receives no estimate and no verdict, because zero cannot be log-transformed and the profiles that would drop out are the low-exposure ones. That metric keeps its row in the table and in the forest plot, with the reason, the zero counts per treatment and the number of profiles that have no value.

Analysing Drug–Drug Interaction Studies

Bioequivalence (Path 6) can also analyse drug–drug interaction (DDI) studies: the design compares two conditions within subjects, with an ANOVA model and a confidence interval for the geometric mean ratio.

How to Set Up a DDI Analysis

Prepare the data as a crossover dataset (Scenario 4 or 5 in Part II). Choose the victim drug given alone as the Reference treatment, so the ratio is (with perpetrator) ÷ (alone): the fold-change in exposure. Upload, map the columns and select the design.

Run the analysis. The app reports the geometric mean ratio and confidence interval for Cmax and AUC. A ratio of 200% means the perpetrator doubled the victim's exposure; 50% means it halved it. Check the Comparison column to confirm which treatment is in the denominator.

Interpreting DDI Results

A DDI study quantifies how large the interaction is; it is not a test of equivalence. ICH M12 (section 5.2) asks for the geometric mean ratio with and without the precipitant and its 90% confidence interval for AUC and C_{max} (and C_{min} when relevant). Interpret the result against no-effect boundaries for the victim drug, preferably based on its exposure-response relationship (section 5.3.1).

- **Default no-effect boundaries:** a 90% CI within 80–125%. M12 calls this generally acceptable but overly conservative for most drugs.

If the study used a sensitive index substrate, the point estimate also classifies the precipitant (M12 section 5.3.2): a strong, moderate or weak inhibitor raises the substrate's AUC at least 5-fold, 2- to 5-fold, or 1.25- to 2-fold; a strong, moderate or weak inducer lowers it by at least 80%, by 50 to 80%, or by 20 to 50%. These classes describe the precipitant, not the size of the interaction for other victim drugs.

Practical Considerations

The app reports the ratio as $\text{Test} \div \text{Reference}$, in percent, with the Reference you chose. A ratio of 350% means the perpetrator increased the victim's AUC 3.5-fold.

For fixed-order DDI studies (all subjects receive the victim alone first, then with the perpetrator), select “Paired comparison”. The app reports the ratio and its confidence interval without a verdict, because period and interaction effects cannot be separated.

The forest plot and confidence interval table can be used in the DDI section of a clinical study report. To compare with no-effect boundaries, enter them as acceptance limits and untick the point-estimate requirement if it does not apply.

Complete Analysis Record

Available from One Subject at a Time, All Subjects and Bioequivalence; Visualize Data produces a similar Figure Record. The zip contains results.xlsx, app_results_reference.csv, analysis_settings.json, nca_pipeline.R (the app's own data-processing code), reproduce_analysis.R, reproduction_check.txt, data_integrity.txt, analysis_summary.html and a copy of the data file. When it creates the record, the app runs the reproduction script and reports whether the result matches (MATCH). data_integrity.txt holds SHA-256 hashes of the data, the settings and the results. A SHA-256 hash is a 64-character fingerprint that changes if a single character in the file changes, so anyone can check later that the files are the ones analysed.

Partial AUC intervals, their roles and the BLQ share used for the notes are recorded, and the reproduction script recalculates the partial AUCs. Change an interval in the settings file afterwards and the check reports DIFFERENT instead of MATCH.

Part IV: Worked Examples

Exercise 1: Theoph Dataset (Batch NCA)

Download the theophylline dataset: open the **Data Preparation Guide** (home screen or top navigation bar) and click **example_theoph.csv** under Example Datasets.

Alternatively, if you have R installed: `write.csv(Theoph, "theoph.csv", row.names = FALSE)`

Upload it. The file has a Dose column in mg/kg; for this exercise do not use it. In All Subjects (Batch) set “Same dose for all subjects”, dose = 320 mg, concentration unit mg/L, route = Oral / IM / SC (extravascular), method = Linear-up / Log-down, and run. “Extravascular” means the drug was not injected into a vein and had to be absorbed first (oral, intramuscular or subcutaneous). Because the fraction absorbed (F) is unknown, clearance and volume are reported as apparent values, CL/F and Vz/F.

Questions: 1) Mean Cmax? 2) Geometric mean AUC to last? 3) Median Tmax? 4) Longest half-life (which subject)? 5) Geometric CV% for AUC ∞ ?

Answers: 1) 8.76 mg/L. 2) 98.7 h·mg/L. 3) 1.14 h. 4) Subject 1 (14.3 h). 5) 28%.

Exercise 2: Bioequivalence

Download the crossover dataset from the **Data Preparation Guide** and click **example_be_crossover.csv**. Upload it and check the mapping. In Bioequivalence set dose = 100 mg, design = 2×2 standard crossover, and check that “Reference” is selected as the Reference treatment. Run.

Questions: 1) Ratio and 90% CI for Cmax? 2) For AUC to last point? 3) Bioequivalent for both? 4) Within-subject CV for Cmax (from the residual variance)?

Answers: 1) 108.62% (106.27–111.02%). 2) 111.25% (108.95–113.60%). 3) Yes, both intervals lie within 80.00–125.00%. 4) About 1.8% ($CV = \sqrt{(\exp(MSE) - 1)}$); the example data are simulated with very little noise.

Exercise 3: Power Calculation

Plan a Study. Find the sample size for standard bioequivalence, 2×2 standard crossover, CV = 25 (percent), expected ratio = 95 (percent), target power = 80 (percent).

Questions: 1) Subjects needed? 2) Power with 30 subjects? 3) Subjects needed at CV = 35%?

Answers: 1) 28 subjects. 2) 83%. 3) 52 subjects.

Exercise 4: Partial AUC

Use the crossover dataset from Exercise 2 (example_be_crossover.csv, sampled at 0, 0.25, 0.5, 1, 1.5, 2, 3, 4, 6, 8, 12 and 24 h). In All Subjects (Batch) add two intervals: 0 to 1 h, with Cmax and Tmax, and 4 h to the last measurable concentration. Then run Bioequivalence with the same intervals, 0–1 h pivotal and 4 h–t supportive, with the log-down method.

Questions: 1) Work out AUC 0–1 h for subject 1 under Test by hand, from the first four samples (0, 12.3, 45.6 and 89.2 ng/mL), and compare it with the table. 2) Ratio and 90% CI

for AUC 0–1 h? 3) What does the table show for AUC 4 h–t, and why? 4) Ratio for C_{max} within 0–1 h?

Answers: 1) 42.475 ng/mL·h ($1.5375 + 7.2375 + 33.7$); the concentrations rise throughout, so both trapezoidal methods give the same value. 2) 113.55% (110.90–116.25%), so bioequivalent. 3) 112.54% (110.10–115.05%) without a verdict, because that interval is supportive. 4) 110.69% (108.31–113.11%).

Part V: Theoretical Background

Trapezoidal Methods for AUC Calculation

The Linear Trapezoidal Rule

Between consecutive observations at t_i and t_{i+1} with concentrations C_i and C_{i+1} , the area is approximated as a trapezoid:

$$\text{AUC}(i) = (t(i+1) - t(i)) \times (C(i) + C(i+1)) / 2$$

This assumes linear concentration change. During elimination, drug concentrations typically decline exponentially, and the linear assumption leads to systematic overestimation: a straight line between two points on an exponential decay lies above the true curve.

The Logarithmic Trapezoidal Rule

If the concentration declines mono-exponentially between two points, the exact integral yields:

$$\text{AUC}(i) = (t(i+1) - t(i)) \times (C(i) - C(i+1)) / (\ln C(i) - \ln C(i+1))$$

This gives a more accurate estimate during the terminal phase. It requires both concentrations to be positive.

The Linear-Up / Log-Down Hybrid

Proposed by Yeh and Kwan (1978): the linear formula while concentrations rise and the logarithmic formula while they fall. It is the default in NCA Assistant.

Terminal Phase Characterisation and Half-Life Estimation

What Is the Terminal Phase?

After distribution is complete, drug elimination follows $C(t) = C_0 \cdot e^{-\lambda_z \cdot t}$. On a log-linear plot, this appears as a straight line with slope $-\lambda_z$. The half-life is $t_{1/2} = \ln(2) / \lambda_z$.

Automated Point Selection

NCA Assistant uses NonCompart's slope selection, the same for the results table and the Half-Life Review. It fits log-linear regressions to the last 3, 4, 5, ... points (after extravascular dosing only points after $C_{\max}$ are used) and takes the fit with the highest adjusted R^2 ; fits within 0.0001 of the best prefer more points. If that fit has an adjusted R^2 below the minimum (slider, default 0.70), the profile gets no half-life, and AUC_∞ , CL/F , Vz/F and MRT are left empty too; a note names the profile. This happens when the terminal phase is poorly sampled, when late samples are near the LLOQ, or when elimination is multi-phasic. A lower minimum accepts noisier fits.

Manual Point Selection

Adjust the points yourself for multi-phasic elimination, flip-flop kinetics, enterohepatic recirculation or late samples near the LLOQ. In Half-Life Review, select a profile, tick the terminal-phase points and click "Recalculate". At least 2 points are needed, and the slope must be negative. NonCompart then recomputes half-life, AUC_∞ , CL/F , Vz/F and MRT from those points, with its unit conversions. A hand-picked selection is used whatever its adjusted R^2 , so judge the fit on the plot. Selections are kept when you run the analysis again and are

listed in the Analysis Record. AUC % Extrapolated is the share of AUC_{∞} that lies beyond the last measured concentration. Values above 20% are flagged in amber: the extrapolation then carries much of the total, so check the terminal phase and consider reporting AUClast as the primary endpoint.

BLQ Handling: Pharmacological Rationale

Every assay has a lower limit of quantification (LLOQ). How BLQ samples are handled changes AUC. Setting early BLQ samples to 0 is usual before absorption; treating late BLQ samples as missing avoids adding area that was never measured. LLOQ/2 is a compromise that adds a little area wherever it is used. Rule 1, the default, is the most common choice in NCA. Rule 6 (LLOQ/2 before the first measurable sample, then zero) suits drugs with an absorption lag. Running two rules and comparing the results shows how much the choice matters. Beal (2001) compares methods for BLQ data in model-based (population) analysis, not in NCA.

Steady-State Considerations

At steady state AUC_{τ} replaces AUC_{∞} . You enter the dosing interval τ ; AUC_{τ} is the AUC from 0 to τ , interpolated between samples and extrapolated with λ_z when the last sample is missing or earlier than τ (not reported if that fit was rejected). $C_{avg} = AUC_{\tau}/\tau$, C_{min} is the lowest concentration between 0 and τ , fluctuation = $(C_{max} - C_{min})/C_{avg} \times 100$ and swing = $(C_{max} - C_{min})/C_{min}$. CL/F and V_z/F are calculated from AUC_{τ} . AUC_{∞} is not meaningful during repeated dosing, and a half-life from one interval is unreliable when the interval does not cover the terminal phase.

Partial AUCs: Interpolation and the End of the Interval

A partial AUC is the area over part of the profile. With both cutoffs on sampling times it is simply the sum of the trapezoids between them.

A cutoff between two samples has to be interpolated: linearly, or log-linearly on a falling segment when you chose linear-up/log-down. The two methods differ most where the curve bends, which is why a protocol that asks for a partial AUC should schedule a sample at the cutoff.

An interval can end at t , the time of the last measurable concentration. That time belongs to the profile, not to the sampling schedule: a later sample that came back below the limit of quantification does not extend it. The app calculates such an interval as AUC_{0-t} minus the AUC from 0 to the start, so an interval ending exactly at the last measurable time gives the same number.

Past that point nothing is reported. Extrapolating with λ_z would turn an observed exposure metric into a model-based one, and it would do so for the profiles that carry the least information.

Part VI: Statistical Methods

The Two One-Sided Tests (TOST) Procedure

Conventional hypothesis testing ($H_0: \mu_T = \mu_R$) can only reject equality, not prove similarity. Schuirmann (1987) reformulated the problem as two one-sided tests:

$$H_{01}: \mu(T) / \mu(R) \leq 0.80 \quad (\text{ratio too low})$$

$$H_{02}: \mu(T) / \mu(R) \geq 1.25 \quad (\text{ratio too high})$$

Both rejected at $\alpha = 0.05 \rightarrow$ ratio lies within $[0.80, 1.25] \rightarrow$ bioequivalent. This is equivalent to: the 90% CI for the ratio falls entirely within $[80\%, 125\%]$. The limits are symmetric on the log scale: $\ln(0.80) = -0.2231$, $\ln(1.25) = +0.2231$.

The ANOVA Model for Crossover Designs

The standard model for a $2 \times 2 \times 2$ crossover:

$$\ln(Y) = \mu + \text{SEQ} + \text{SUBJ}(\text{SEQ}) + \text{PER} + \text{TRT} + \varepsilon$$

SEQ accounts for differences between sequence groups, SUBJ removes between-subject variability from the residual, PER accounts for changes between periods, and TRT is the effect of interest: its coefficient estimates the log-ratio of geometric means. NCA Assistant fits this model for all crossover designs, with Sequence, Subject, Period and Treatment as factors; with fixed effects this is EMA Method A.

Fixed-Effects vs. Mixed-Effects Models

Fixed effects (lm): all terms, including Subject, are fixed (EMA Method A). The EMA guideline asks for fixed effects. Subjects without both treatments do not contribute to the comparison (complete-case analysis). Residual df = $N - 2$ for a balanced $2 \times 2 \times 2$.

Mixed effects (nlme::lme): Subject is a random intercept. Subjects with only one treatment still contribute, which can help with dropouts. Denominator df use the containment method. Use the model your protocol or statistical analysis plan specifies. If the mixed model cannot be fitted, the app uses fixed effects and says so in the Model column. This is EMA Method B (one within-subject variance; results agree with replicateBE). It is not the FDA model for replicate designs, which has separate Test and Reference variances, a subject-by-formulation interaction and Satterthwaite degrees of freedom.

Confidence Interval Construction

$$90\% \text{ CI} = \exp(\delta \pm t(0.95, \text{df}) \times \text{SE}(\delta)) \times 100\%$$

where δ is the treatment coefficient on the log scale, SE its standard error from the model, and df the denominator degrees of freedom. The limits are rounded to two decimals before comparison with the acceptance limits (FDA, 2026). Bioequivalence is concluded when the rounded interval lies within the limits (80.00–125.00% by default) and, for wider limits, the point estimate lies within 80.00–125.00%.

Design-Specific Models

Paired comparison: $\ln(Y) = \mu + \text{SUBJ} + \text{TRT} + \varepsilon$, equivalent to a paired t-test; $\text{df} = N - 1$. Ratio and interval only, no verdict.

Parallel design: $\ln(Y) = \mu + \text{TRT} + \varepsilon$. $df = N_T + N_R - 2$. Differences between subjects enter the comparison, so more subjects are needed than in a crossover. The interval assumes equal variances in both groups (pooled variance).

Replicate designs: the same model, with Period at 3 or 4 levels; each administration enters once. The confidence interval uses average bioequivalence limits. The within-subject variability table uses a separate model per treatment (Sequence, Subject and Period, fitted to subjects with that treatment at least twice), as in the replicateBE package; the app's results agree with replicateBE on its 30 reference data sets.

Drug–Drug Interaction Analysis

DDI studies use the same statistics as BE: the ANOVA model, the geometric mean ratio and its 90% confidence interval. The difference is the interpretation: the ratio is the fold-change in exposure caused by the interacting drug, judged against no-effect boundaries for the victim drug (see Part III) rather than BE acceptance limits.

ICH M12 on drug interaction studies (2024), published as guidance by EMA and FDA, asks for the geometric mean ratio of AUC and C_{max} (and C_{min} when relevant) with and without the precipitant, with its 90% confidence interval (section 5.2). Crossover studies, including one-sequence studies, are preferred over parallel designs (section 3.2.1).

Power and Sample Size for Bioequivalence

Exact power via Owen's Q function for standard ABE. For ABEL, RSABE and NTID, power is simulated in PowerTOST: 100,000 simulated studies for the sample size, 10,000 for power and the power curve. Inputs: within-subject CV (for scaled methods, of Test and Reference), expected ratio, target power and design. A higher CV or a ratio further from 100% needs more subjects; a crossover needs fewer subjects than a parallel design.

Practical guidance: without data, a CV of 30% is a cautious assumption. The expected ratio defaults to 95% for standard bioequivalence, 90% for highly variable drugs and 97.5% for narrow therapeutic index drugs; target power 80%, or 90% for studies where failure is costly. Check the sensitivity plot to see how N changes with the CV.

Partial AUCs in Bioequivalence

Regulators ask for a partial AUC when two products can share C_{max} and total AUC while the shape of the curve differs. FDA product-specific guidances name the metric and the interval per product; Hopefl et al. (2025) list the guidances published between 2020 and 2024.

The statistics do not change. A partial AUC is log-transformed and analysed with the same model and confidence interval as C_{max} and AUC. The interval comes from the protocol; choosing it after seeing the data inflates the chance of a false conclusion of equivalence.

A pivotal metric has to meet the acceptance limits, a supportive one is reported without a verdict. With several pivotal metrics each has to pass on its own. That is an intersection-union test, so no correction for multiple testing applies.

Zeros need care. An early interval with only BLQ samples can be exactly 0, which has no logarithm. Leaving those profiles out would remove the lowest exposures and push the ratio

up, so the app reports no estimate and no verdict for that metric and shows how many zero values each treatment had.

For every metric the table also counts, per treatment, the profiles that had no value, the profiles with a value of zero, and the profiles resting mainly on concentrations set by the BLQ rule. The last count matters because such a profile can pull a ratio on the log scale without ever being zero: under Rule 1 an early window may be exactly zero and get no verdict, while under Rule 6 the same sample becomes LLOQ/2 and the metric keeps a verdict that one imputed value helped to decide.

The EMA allows the acceptance criteria for a partial AUC to be widened in the same way as for C_{max} of a highly variable product (EMA/CHMP/EWP/280/96 Rev1, section 6.8.2.2). The app does not scale limits: for a replicate design it shows CV_{wR} and the limits it would imply, and it applies the limits you enter. The same guideline recommends an early and a terminal partial AUC, split at a pre-specified cut-off such as half the dosing interval, when a single-dose study replaces a multiple-dose study.

Plan such a study with the CV of the partial AUC itself. For loxapine inhalation powder the within-subject CV was above 30% for AUC 0–10 min but below 30% for AUC 0–30 min, which is why the 30-minute window became the pivotal metric (Hopefl et al., 2025).

Regulatory Guidelines

EMA Guideline on Bioequivalence (CPMP/EWP/QWP/1401/98 Rev. 1, 2010):

<https://www.ema.europa.eu/en/investigation-bioequivalence-scientific-guideline>

FDA Statistical Approaches to Establishing Bioequivalence (May 2026; replaces the 2001 guidance): <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/statistical-approaches-establishing-bioequivalence>

FDA Bioavailability Studies Submitted in NDAs or INDs: General Considerations

(2022; replaces the 2003 guidance): <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/bioavailability-studies-submitted-ndas-or-ind-s-general-considerations>

ICH M12 Drug Interaction Studies (2024; published by EMA and FDA):

<https://www.ema.europa.eu/en/ich-m12-drug-interaction-studies-scientific-guideline>

CDISC SDTM Controlled Terminology (release 2026-03-27, source of the parameter codes): <https://evs.nci.nih.gov/ftp1/CDISC/SDTM/>

Appendices

Appendix A: Glossary

Abbreviation	Definition
ABE	Average Bioequivalence (80–125% limits by default)
ABEL	Average Bioequivalence with Expanding Limits (EMA, for HVDs)
ADNCA	CDISC ADaM dataset for non-compartmental analysis
ANOVA	Analysis of Variance
AUC	Area Under the Curve (total drug exposure)
AUC _T	AUC within the dosing interval (steady state)
BLQ	Below the Limit of Quantification
CL/F	Apparent clearance
C _{max}	Maximum observed concentration
CV%	Coefficient of Variation
CV _{wR} / CV _{wT}	Within-subject coefficient of variation of the Reference / Test
DDI	Drug–Drug Interaction
GMR	Geometric Mean Ratio (Test/Reference)
HVD	Highly Variable Drug (within-subject CV of the Reference above 30%)
LLOQ	Lower Limit of Quantification
MRT	Mean Residence Time
NCA	Non-Compartmental Analysis
NTID	Narrow Therapeutic Index Drug
PE	Point estimate (geometric mean ratio)
Reference	The treatment in the denominator of the ratio, chosen by the user
RSABE	Reference-Scaled Average Bioequivalence (FDA, for HVDs)
T _{max}	Time of maximum observed concentration
TOST	Two One-Sided Tests
t _{1/2}	Terminal elimination half-life
V _z /F	Apparent volume of distribution (terminal phase)
λ _z	Terminal elimination rate constant
Partial AUC	Area under the curve over part of the profile, for example AUC 0–30 min. The interval comes from the protocol or a product-specific guidance.
Pivotal metric	A bioequivalence metric that has to meet the acceptance limits. It receives a verdict.
Supportive metric	A bioequivalence metric reported with its ratio and confidence interval, without a verdict.
t (last measurable)	The time of the last measurable concentration of a profile. An interval written as 4 h–t ends there.

Appendix B: Recognised Column Names

Meaning	Recognised names (case-insensitive)
Subject	Subject, SubjectID, USUBJID, ID, Pat, PatID, Proband, Teilnehmer
Time	Time, Hours, Hour, Tpt, NTIM, APTS, Zeit, Tid
Concentration	Conc, Concentration, DV, Cp, Result, Konz, Plasma, or a name containing ng/mL or ug/L
Treatment	Treatment, Trt, Formulation, Drug, Arm, Behandlung
Period	Period, Per, Prd, Phase
Sequence	Sequence, Seq, Grp
Dose	Dose, Amt, Amount, Dosis

Names are not case-sensitive. When no name matches, the app falls back to a column by position, so always check the suggested mapping.

Appendix C: Troubleshooting

“Disconnected from the server”: the analysis exceeded memory or time limits, or the app met an unexpected error. Run the app locally in RStudio and check the console.

“Error: An error has occurred”: an unexpected error in a calculation or plot. Check the LLOQ, BLQ rule, units and column mapping, and report the problem with a description of the data if it persists.

Half-life is empty: the automatic fit is below the minimum adjusted R^2 . Review the profile in Half-Life Review, or lower the minimum if that is justified.

Half-life differs from other NCA software: check the minimum R^2 , the point selection in Half-Life Review, the BLQ rule and the trapezoidal method; programs differ in how they choose the terminal-phase points.

Comma decimals not recognised: set “Decimal point” to “Comma (,)” (and the delimiter, often semicolon) in the upload settings.

Large sampling gap warning: for studies longer than 48 h the threshold is 48 h, so daily trough samples are not flagged; for shorter studies it is 24 h. A gap close to the threshold is usually normal sampling.

“Profile(s) do not start near time zero”: time is probably measured from the first dose of the study. Use the time since the dose of each period (in CDISC terms ARRLT or NRRLT, not AFRLT).

“Choose the Reference treatment” the run is blocked until you choose the Reference treatment in the Bioequivalence settings. The app pre-selects it only for unambiguous names (R, Ref, Reference, Comparator, Innovator, Originator, RLD).

Unit error before the NCA run the combination of dose, time and concentration units is not supported, or molar units are combined with a mass dose without a molecular weight. Select supported units and enter the molecular weight.

“Refused: ...” in the CDISC ADNCA upload, the message says why (for example derived records, several units, AFRLT as time, or AVAL missing without a BLQ result). Fix the dataset upstream or make the choice the message asks for.

My partial AUC is empty: the interval reaches past the last measurable concentration of that profile, or starts after it. The note above the results names the profile and its last measurable time. Partial AUCs are not extrapolated.

No verdict for my partial AUC: the interval is marked supportive, or the metric has a value of zero in at least one profile. The table gives the reason and the number of zero values per treatment.

Appendix D: References

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